

ESG is the Most Polarizing Nonwage Amenity: Evidence from a Field Experiment in Brazil

By EMANUELE COLONNELLI, TIM MCQUADE, GABRIEL RAMOS, THOMAS RAUTER, AND OLIVIA XIONG*

* Colonnelli: University of Chicago Booth School of Business and NBER, CEPR, BREAD, J-PAL (email: emanuele.colonnelli@chicagobooth.edu). McQuade: University of California, Berkeley Haas School of Business and NBER (email: tmcquade@haas.berkeley.edu). Ramos: Imperial College London (email: g.lobato-ramos23@imperial.ac.uk). Rauter: University of Chicago Booth School of Business (email: thomas.rauter@chicagobooth.edu). Xiong: University of Chicago Booth School of Business (email: olivia.xiong@chicagobooth.edu).

Over the past decade, corporations have faced increasing pressure to prioritize not only shareholder interests, but also broader societal objectives, such as climate change or various social issues. This shift has brought greater attention to firms' environmental, social, and governance (ESG) practices (Hart and Zingales 2017). At the same time, academic research has increasingly focused on how individuals consider nonmonetary job factors when making employment decisions (e.g., Cassar and Meier 2018; Mas and Pallais 2020).

In Colonnelli et al. (2025), we investigate the role of ESG as a nonwage amenity for talent allocation in Brazil by combining a field experiment with administrative matched employee-employer microdata to estimate an

equilibrium model of the labor market. We find that job-seekers, on average, exhibit a strong preference to work for firms with good ESG practices. However, these preferences are highly heterogeneous across the sociodemographic spectrum, with the strongest ESG preferences observed among highly educated, white, and politically liberal individuals.

The finding that firms' ESG practices seem to polarize job-seekers along multiple sociodemographic dimensions echoes the growing academic literature showing that political and social identities shape opinions about ESG.¹ Prior studies have shown that political polarization affects corporate behavior, investment decisions, and household financial and consumer choices (e.g., Hong and Kostovetsky 2012; Colonnelli et al. 2024a; Colonnelli et al. 2024b; Conway and Boxell 2024; Fos et al. 2024; McCartney et al. 2024; for a review, see Kempf and Tsoutsoura 2024). The polarizing nature of ESG is further

¹ Throughout this paper, we use the term polarization broadly to refer to the divergence of preferences among distinct demographic groups.

consistent with recent studies documenting differences in preferences or engagement by educational attainment or race (e.g., Schuldt and Pearson 2016; Bauer et al. 2021; Seifert et al. 2024).

It remains an open question whether ESG is a uniquely polarizing nonwage amenity compared to others. A broad range of nonwage amenities exists, and preferences for these amenities may also be polarizing across various socioeconomic and political dimensions. For example, remote work, which is a key determinant of individuals' workplace preferences (including in our study), may exhibit differential importance for several different subgroups of the population.

Our experimental framework provides an ideal setting to formally investigate this question, as our randomized design ensures orthogonality among job attributes and allows us to isolate preferences for each nonwage amenity in an unbiased way. In this paper, we extend our main analysis to explore the heterogeneity in preferences for a broad range of nonwage amenities commonly observed in the real world and studied in the labor literature (Mas and Pallais 2020).

I. Experimental Setting

A. The Job Rating Experiment—

To estimate job-seekers' preferences for nonwage amenities, we adopt a method similar to the nondeceptive incentivized resume rating approach by Kessler et al. (2019). Job-seekers rate their interest in a series of synthetic job postings constructed by our research team, with components such as wages, non-wage amenities, and other firm and job characteristics randomized by design.

Our experimental survey begins with a set of “filtering” questions regarding respondents' education level and preferred professional areas. We use their responses to these questions to ensure that participants are only shown job postings relevant to their qualifications and interests. Participants then evaluate their interest in 20 unique, synthetic job postings, which we detail further in the following subsection. We conclude with questions on participants' sociodemographic characteristics. Our main paper, Colonnelli et al. (2025), provides all additional relevant details.

To implement the experiment, we collaborate with Catho, the leading job matching platform in Latin America. Catho invited its active members to participate in the survey between September and November 2022, resulting in a final sample of 1,206 responses.

Our experimental design has several key advantages. First, job-seekers are informed that the job postings are synthetic and created by our research team, eliminating any deception and reducing the risk of biases that commonly arise in survey-based experiments. Second, participants are incentivized to respond truthfully: we inform respondents that their evaluations will feed into a machine learning algorithm that will match them to real job postings aligned with their preferences. Finally, our design allows us complete flexibility in creating the job postings. This ensures that job attributes are independent of each other, enabling us to cleanly isolate preferences for specific job features while controlling for confounding factors.

B. Synthetic Job Postings—

Creating Job Postings.—To construct the synthetic job postings, we systematically reviewed a large set of randomly selected real job postings from Catho’s platform. We identified the key components of typical job postings—such as job descriptions, qualifications, firm attributes, wages, and nonwage amenities—and analyzed their content and visual layout to ensure the synthetic postings accurately represent real job postings in Brazil. We then approximate the probability distribution of each component’s

inclusion in real postings. To generate synthetic job postings, we randomize components based on their inclusion probabilities. We provide a sample job posting in Figure 1.

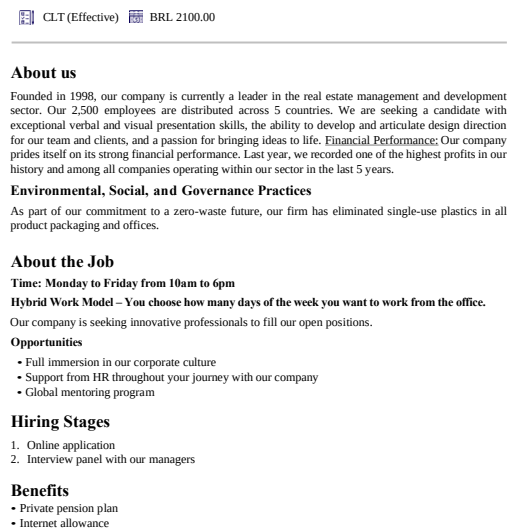


FIGURE 1. EXAMPLE OF SYNTHETIC JOB POSTING

Note: This figure provides an example of a synthetic job posting used in the experiment.

Nonwage Amenities.—Nonwage amenities, defined as job attributes that might provide value to employees beyond direct wages, are a key component of the job postings. We incorporate standard nonwage amenities commonly observed in real-world settings and extensively studied in the labor literature, such as transportation subsidies, meal or food allowances, private pension plans, training programs, and remote work arrangements, among others.

ESG as a Nonwage Amenity.—ESG attributes are featured in a subset of job

postings as a distinct category of nonwage amenities. These attributes highlight a firm's ESG efforts in specific practices, including: (i) *environmental* practices (e.g., recycling, land use, energy efficiency); (ii) *social* practices (e.g., diversity & inclusion, professional development); and (iii) *governance* practices (e.g., policies against lobbying, bribery, and corruption). At least one ESG signaling sentence was included with an unconditional probability of 26%. We also show ESG certifications in a subset of job postings.

Primary Job Characteristics.—At the top of each job posting, we include three core components: (i) job title, (ii) location, and (iii) wage. Respondents are shown relevant job titles and locations based on their answers to the filtering questions. To ensure realism, wages are drawn from distributions tailored to respondents' education level and preferred professional area.

Other Job Attributes.—In addition to benefits, ESG practices, and primary job characteristics, the synthetic job postings include several additional randomized job attributes, such as the firm's characteristics (e.g., number of employees, company age, financial strength), on-the-job opportunities and activities, work-from-home arrangements, and hiring stages. For additional details on

these components, please refer to Colonnelli et al. (2025).

C. Estimating Preferences for Job Attributes—

To estimate job-seekers' preferences for specific job and firm characteristics, we ask participants to rate their interest in each job posting by responding to the following question: “*How interested would you be in receiving an offer for this job position?*”

The response to this question, which we refer to as *Interest*, is our main dependent variable. *Interest* is measured on a 7-point Likert scale, with higher scores indicating stronger interest in the job.

A potential confounding factor in interpreting these preferences is that job-seekers' interest in a job might be influenced by their perception of the likelihood of receiving an offer. To address this concern, we explicitly instructed respondents: “Imagine that the employer guarantees you a job offer—consider only your perception of the quality of the position.” This framing ensures that the reported interest reflects intrinsic preferences for the job attributes, independent of hiring expectations.

II. Estimation & Results

To examine whether ESG is uniquely polarizing, we compare the heterogeneity of

preferences for ESG to those of other nonwage amenities. Specifically, we analyze how preferences for each amenity vary across key sociodemographic groups.

First, we categorize job-seekers into binary groups based on their sociodemographic characteristics: education (equal to one for individuals with a college degree or higher); race (equal to one for individuals identifying as white); political views (equal to one for individuals identifying as liberal or moderate); age (equal to one for individuals aged 45 years old or younger); and gender (equal to one for individuals identifying as female). We then interact each nonwage amenity with the corresponding binary indicator.

We estimate specifications of the following form:

$$(1) \text{Interest}_{ij} = \alpha + \beta_0(NWA_{ij} \times SDC_i) + \beta_1 SDC_i + \beta_2 NWA_{ij} + \beta_3 \ln(Wage_{ij}) + \beta_4 FS_{ij} + \beta_5 \text{Number of NWA}_{ij} + \text{StrataFE} + \text{IndividualFE} + \epsilon_{ij},$$

where i represents the i^{th} respondent, and j indicates the job posting evaluated by respondent i . *Interest* captures the level of interest a respondent has in a job posting. *SDC* is a binary indicator denoting a given sociodemographic characteristic. *NWA* is a binary indicator equal to one if the job posting

displays a given nonwage amenity. $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting, *FS* is a binary indicator equal to one if the job posting signals the company's financial strength, and *Number of NWA* is the number of nonwage amenities shown in the given job posting. We also include strata fixed effects, defined as binary indicators for each combination of a respondent's education level and the professional area they are interested in working in, as well as individual fixed effects.

[Table 1 Placeholder]

Table 1 presents the estimated coefficients for the interaction terms $NWA \times SDC$, as specified in Equation 1. Each coefficient reported in the table is derived from a separate regression, where a specific nonwage amenity (listed in the table rows) is interacted with a particular sociodemographic characteristic (shown in the table columns). For example, the coefficient of 0.114 in the first column and first row represents the interaction between the nonwage amenity *ESG Practices* and the sociodemographic characteristic *High Education*, capturing the preference highly educated respondents have for ESG practices relative to respondents without a college degree. Similarly, the coefficient of 0.078 in the second column and first row reflects the

interaction between the nonwage amenity *ESG Practices* and the sociodemographic characteristic *White*, capturing the preference white respondents have for ESG practices relative to nonwhite respondents.

As established in our companion paper, the coefficient on *ESG Practices* (a binary indicator equal to one if the job posting contains at least one ESG signaling sentence or third-party ESG certification) is positive and statistically significant at the 95% level or higher in Columns (1) to (3), indicating that preferences for ESG are stronger among individuals who are highly educated, white, and self-identify as politically liberal or moderate. These interaction coefficients range from 0.078 (for white individuals; Column 2) to 0.114 (highly educated individuals; Column 1). In terms of economic magnitude, a coefficient estimate of 0.1 corresponds to a wage equivalent of BRL 390, or 9.33% of the average wage in the sample.

When we next break down *ESG Practices* into subcomponents, we find significant heterogeneity in preferences for *Environmental Practices* and *Governance Practices*, but not for *Social Practices*. Preferences for *Environmental Practices* are strongest among white respondents and those who are politically liberal or moderate, with coefficients for both groups statistically significant at the 95% level.

Highly educated individuals exhibit the strongest preferences for *Governance Practices*, with the coefficient statistically significant at the 99% level.

Among all nonwage amenities considered in this study, ESG practices emerge as the most polarizing. Of the 17 nonwage amenities analyzed, 14 exhibit no meaningful heterogeneity across any sociodemographic dimension. This includes important nonwage amenities such as private pension plans, various medical, educational, and personal assistance, and mentoring and training programs, among others, which show relatively uniform preferences across all groups.

The only exceptions include *Work-from-Home* policies, a key nonwage amenity that drives the highest average interest among job-seekers (Colonnelli et al. 2025), and *Personal Development* programs. The coefficient for *Work-from-Home* is positive and statistically significant at the 99% level in Column (5). This finding aligns with previous studies showing that women value *Work-from-Home* more than men (e.g., Mas and Pallais 2017; Barrero et al. 2021; Goldin 2022; Bloom et al. 2024), which also validates our experimental design. Moreover, our results on *Work-from-Home* are in line with the findings by Wiswall and Zafar (2018) and Le Barbanchon et al. (2021) that women have stronger preferences for

flexibility and shorter commute times in job arrangements. Similarly, for *Personal Development* programs, we observe that highly educated job-seekers place greater value on this amenity compared to those without a college degree, although this difference is statistically significant only at the 90% level.

However, even for these two nonwage amenities, heterogeneity is limited to a single sociodemographic dimension—gender for *Work-from-Home*, and education for *Personal Development*. In contrast, ESG preferences exhibit far greater degrees of heterogeneity across multiple sociodemographic dimensions, including education, race, and political affiliation. Overall, ESG appears to be the most divisive of all nonwage amenities, setting it apart from other job attributes.

IV. Conclusion

Using an incentivized field experiment with real job seekers in Brazil, we find that ESG is both highly valued and uniquely polarizing as a nonwage amenity. Extending our analysis to a broad range of nonwage amenities commonly observed in the real world and studied in the labor literature, we show that this polarizing nature is not prevalent among other amenities.

The polarization of ESG underscores the importance of understanding the distributional consequences of nonwage amenities and the

growing role of corporate culture in shaping job-seeker preferences.

REFERENCES

- Barrero, Jose Maria, Nicholas Bloom, and Steven J. Davis.** 2021. “Why Working from Home will Stick.” *NBER Working Paper No. 28731*.
- Bauer, Rob, Thobas Ruof, and Paul Smeets.** 2021. “Get Real! Individuals Prefer More Sustainable Investments.” *The Review of Financial Studies* 34(8): 3976-4043.
- Bloom, Nicholas, Ruobing Han, and James Liang.** “Hybrid Working from Home Improves Retention Without Damaging Performance.” *Nature* 630: 920-925.
- Cassar, Lea, and Stephan Meier.** 2018. “Nonmonetary Incentives and the Implications of Work as a Source of Meaning.” *Journal of Economic Perspectives* 32(3): 215-238.
- Colonnelli, Emanuele, Tim McQuade, Gabriel Ramos, Thomas Rauter, and Olivia Xiong.** 2025. “Polarizing Corporations: Does Talent Flow to ‘Good’ Firms?” *Available at SSRN* 4645695.
- Colonnelli, Emanuele, Niels J. Gormsen, and Tim McQuade.** 2024a. “Selfish Corporations.” *Review of Economic Studies* 91(3): 1498-1536.
- Colonnelli, Emanuele, Valdemar P. Neto,**

- and Edoardo Teso.** 2024b. "Politics at Work." *NBER Working Paper No. 30182*.
- Conway, Jacob, and Levi Boxell.** 2024. "Consuming Values." *Available at SSRN* 4855718.
- Fos, Vyacheslav, Elisabeth Kempf, and Margarita Tsoutsoura.** 2022. "The Political Polarization of Corporate America." *NBER Working Paper No. 30183*.
- Goldin, Claudia.** 2022. "Understanding the Economic Impact of COVID-19 on Women." *Brookings Papers on Economic Activity* 2022(1): 65-139.
- Hart, Oliver and Luigi Zingales.** 2017. "Companies Should Maximize Shareholder Welfare Not Market Value." *Journal of Law, Finance, and Accounting* 2: 247-275.
- Hong, Harrison, and Leonard Kostovetsky.** 2012. "Red and Blue Investing: Values and Finance." *Journal of Financial Economics* 103(1): 1-19.
- Kempf, Elisabeth, and Margarita Tsoutsoura.** 2024. "Political Polarization and Finance." *Annual Review of Financial Economics* 16.
- Kessler, Judd B., Corrine Low, and Colin D. Sullivan.** 2019. "Incentivized Resume Rating: Eliciting Employer Preferences Without Deception." *American Economic Review* 109(11): 3713-3744.
- Le Barbanchon, Thomas, Roland Rathelot, and Alexandra Roulet.** "Gender Differences in Job Search: Trading Off Commute Against Wage." *The Quarterly Journal of Economics* 136(1): 381-426.
- Mas, Alexandre, and Amanda Pallais.** 2020. "Alternative Work Arrangements." *Annual Review of Economics* 12(1): 631-658.
- Mas, Alexandre, and Amanda Pallais.** 2017. "Valuing Alternative Work Arrangements." *American Economic Review* 107(12): 3722-3759.
- McCartney, W. Ben, John Orellana-Li, and Calvin Zhang.** 2024. "Political Polarization Affects Households' Financial Decisions: Evidence from Home Sales." *The Journal of Finance* 79(2): 795-8.
- Schuldt, Jonathon P., and Adam R. Pearson.** 2016. "The Role of Race and Ethnicity in Climate Change Polarization: Evidence from a US National Survey Experiment." *Climatic Change* 136: 495-505.
- Seifert, Marcel, Florian Spitzer, Simone Haeckl, Alexia Gaudeul, Erich Kirchler, Stefan Palan, and Katharina Gangl.** 2024. "Can Information Provision and Preference Elicitation Promote ESG Investments? Evidence from a Large, Incentivized Online Experiment." *Journal of Banking & Finance* 161: 107114.
- Wiswall, Matthew, and Basit Zafar.** 2018. "Preference for the Workplace, Investment in

Human Capital, and Gender.” *The Quarterly Journal of Economics* 133(1): 457-507.

TABLE 1— PREFERENCES FOR ESG PRACTICES AND NONWAGE AMENITIES ACROSS SOCIODEMOGRAPHIC GROUPS

	High Education	White	Liberal or Moderate	Young	Female
	(1)	(2)	(3)	(4)	(5)
ESG Practices	0.114***	0.078**	0.093**	-0.047	-0.015
Environmental Practices	0.075	0.108**	0.118**	-0.045	-0.005
Social Practices	0.121	0.030	-0.114	-0.055	0.024
Governance Practices	0.169**	0.039	0.087	-0.084	0.034
Daycare Allowance	-0.066	-0.024	-0.045	0.013	0.012
Gym and Physical Activity Support	-0.033	0.007	-0.038	-0.014	-0.024
Internet Allowance	0.072	-0.017	-0.019	0.006	0.019
Language Course Allowance	0.036	0.056	0.050	-0.005	-0.016
Meal and Food Allowance	-0.007	-0.030	-0.025	0.029	0.058
Medical, Educational, and Personal Assistance	-0.027	0.046	-0.039	-0.004	-0.002
Mentoring and Training Program	0.016	0.006	0.028	0.016	-0.044
Multinational	-0.003	-0.042	0.002	0.036	0.025
Partnerships with Educational Organizations	0.018	-0.040	0.066	-0.042	0.007
Personal Development	0.062*	-0.016	0.058	0.052	-0.018
Pharmacy Allowance	0.067	0.015	-0.035	-0.022	-0.039
Pregnancy Allowance	0.015	-0.073	0.027	0.013	0.051
Private Pension Plan	0.002	0.047	-0.018	0.057	0.070
Transportation Allowance	0.008	0.014	0.043	0.046	-0.058
Wellness Program	-0.016	-0.018	0.027	0.027	-0.043
Work-from-Home	0.046	-0.051	-0.019	-0.014	0.124***



Negative and Statistically Significant



Not Statistically Significant



Positive and Statistically Significant

Notes: This table presents the interaction coefficients (β_0) from separate regressions for each combination of sociodemographic characteristic and nonwage amenity using the following regression specification: $Interest_{ij} = \alpha + \beta_0(NWA_{ij} \times SDC_i) + \beta_1SDC_i + \beta_2NWA_{ij} + \beta_3\ln(Wage_{ij}) + \beta_4FS_{ij} + \beta_5Number\ of\ NWA_{ij} + StrataFE + IndividualFE + \epsilon_{ij}$, where i represents the i^{th} individual, and j represents the job posting rated by individual i . The variable SDC is a sociodemographic characteristic indicator that equals one for the following groups: in Column (1), respondents with a college degree; in Column (2), respondents who are white; in Column (3), respondents who self-identify as politically liberal or moderate; in Column (4), respondents aged 45 years or younger; and in Column (5), respondents who are female. NWA is an indicator equal to one if the job posting displays the nonwage amenity in the lefthand column. $\ln(Wage)$ is the natural logarithm of the monthly wage displayed in the job posting, FS is an indicator equal to one if the job posting signals the company's financial strength, and $Number\ of\ NWA$ is the number of nonwage amenities shown in the job posting. We include strata fixed effects, which are binary indicators for each combination of respondent education level and preferred professional area, and individual fixed effects. Robust standard errors are reported in parentheses. Statistical significance is denoted as follows: * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$.